

Long-term survival and colony growth of *Acropora palmata* fragments transplanted by volunteers for restoration

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ABSTRACT

1. Many branching corals are fragmented by storms, which can serve as a mechanism of asexual reproduction for species that are able to reattach themselves to the substratum and form new colonies. Fragments can also be manually reattached as a means of reef restoration.

2. The growth and survival of 832 fragments of Elkhorn coral, *Acropora palmata*, that were transplanted for a restoration project in the British Virgin Islands was modelled.

3. Mortality was higher in the first year after transplanting than in subsequent years, perhaps reflecting stress from handling or failure of the attachment method.

4. Survival also varied with the year of transplantation (from 2005–2011), and was lowest in years with major storms (2007 and 2010).

5. Fragment survival increased with increasing initial size, with the largest fragments (surface area roughly 1000 cm²) faring substantially better than the smallest (roughly 10 cm²) and average sized fragments (roughly 100 cm²).

6. Colony size (surface area of live tissue) tended to decrease slightly in the first 3 months after being reattached, presumably due to stress from transplanting. Subsequently, the surface area of surviving colonies tended to progressively increase over time, with fragments typically reaching 3000 cm² after 7 years. Colony growth was, however, extremely variable and largely independent of initial colony size.

7. Despite initial reductions in growth and survival due to transplanting, long-term survival of transplanted fragments was roughly comparable with that of natural colonies. Transplanting fragments is thus a promising tool for grass-roots restoration projects.

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INTRODUCTION

As corals have progressively declined in abundance worldwide over the past 30 years (Gardner *et al.*, 2003; Bellwood *et al.*, 2004), there has been an increasing interest in developing methods to restore coral populations (Harriott and Fisk, 1988; Edwards and Clark, 1998; Rinkevich, 2005, 2008; Precht, 2006). One approach to restoration is seeding broken fragments from coral colonies on damaged reefs, or manually reattaching fragments to the substratum. Coral colonies fragment naturally through the actions of storms, bioerosion and predation, but fragmentation also occurs when boats, divers and snorkelers collide with the reef (Woodley *et al.*, 1981; Rogers *et al.*, 1982; Bruckner and Bruckner, 2001; Hawkins *et al.*, 2005). Fragments used for restoration include those generated naturally by storms, by unintentional human impact, and also created deliberately by pruning natural colonies (Rinkevich, 2005; Precht, 2006).

The process of fragmentation is important to the natural demography of branching scleractinian corals, as well as other branching colonial invertebrates on coral reefs (Lasker, 1984; Wallace, 1985; Karlson, 1986; Lewis, 1991). Under favourable conditions, coral fragments may naturally reattach to the reef and form new colonies (Smith and Hughes, 1999). This process of fragmentation and reattachment is hypothesized to be an adaptive strategy for branching corals to colonize new habitats, spread the risk of mortality, and circumvent mechanical limits on colony size (Highsmith, 1982; Wallace, 1985).

For natural coral populations, population dynamic models based on individual body size are often more accurate than are models based on age (Hughes, 1984). Because individual corals are colonies of genetically identical modules, the importance of colony size derives in part from the fact that colony growth is indeterminate and that colonies can also shrink through the death of some modules (partial mortality) (Connell, 1973). Analyses of growth and survival of naturally generated fragments have focused on the fragments created in enormous numbers after major storms. The fate of these fragments is independent of the parent colony (Highsmith, 1982) and their chance of

survival is typically very low (Tunnicliffe, 1981; Fong and Lirman, 1995; Smith and Hughes, 1999). In some cases, larger fragments had an improved chance of post-storm survival (Loya, 1976; Highsmith *et al.*, 1980), but others found no effects of size (Lirman and Fong, 1997), or observed size-distributions consistent with better survival of smaller fragments (Rogers *et al.*, 1982).

A comparable understanding of the demography of transplanted fragments is valuable to assess the success of restoration projects. Variable effects of fragment size on survival have been observed for artificially seeded fragments that were loose on the substratum, with examples reported of size-dependence (Heyward and Collins, 1985; Bowden-Kerby, 1997, 2001; Smith and Hughes, 1999) and size-independence (Bruno, 1998; Bowden-Kerby, 2001; Lindahl, 2003). The fate of fragments that have been manually re-attached to the substratum also appears to be size-dependent in some cases (Bowden-Kerby, 2001; Bruckner and Bruckner, 2001; Raymundo and Maypa, 2004; Okubo *et al.*, 2005, 2007, 2009; Garrison and Ward, 2008) but not others (Kobayashi, 1984; Yap *et al.*, 1998; Bowden-Kerby, 2001; Lindahl, 2003). The survival and colony growth of fragments used for restoration can also be reduced by the handling and disturbance involved in the transplanting process (Yap and Gomez, 1985; Plucer-Rosario and Randall, 1987; Yap *et al.*, 1992; Forrester *et al.*, 2011) and survival may also be reduced by the failure of methods used to reattach the fragment to the reef (Borneman and Lowrie, 2001; Bruckner and Bruckner, 2001).

Few studies have compared the relative effects of natural and transplant-related influences on the demography of transplanted coral fragments. The objectives of this study were thus to test the influence on colony growth and survival of: (1) the year when fragments were transplanted; (2) the time elapsed since transplanting; and (3) the size of fragments. Differences in colony growth and survival among years should reflect changes in natural conditions (e.g. storms, temperature anomalies) that influence coral demography. Effects of elapsed time are hypothesized because stress from handling fragments and failure of the attachment procedure should reduce survival immediately after transplanting.

Because most experimental transplanting studies are of relatively short duration (1 year or less), fragments were tracked over 7 years in order to assess the longer-term fate of transplanted corals (Harriott and Fisk, 1988; Edwards and Clark, 1998; Rinkevich, 2005; Precht, 2006).

The study species was the Elkhorn coral, *Acropora palmata* (Lamarck, 1816). *Acropora palmata* is a major reef-building coral in the Caribbean, and was formerly the dominant coral species in shallow wave-exposed areas (Goreau, 1959). During the 1980s and 1990s, *A. palmata* declined by 80–98% (Precht *et al.*, 2004), making it a priority candidate for restoration and prompting its listing on the US Endangered Species Act, the IUCN red list and CITES Appendix II. *Acropora palmata* has a branching morphology and reproduces both asexually, by fragmentation, and sexually (Bak and Engel, 1979; Dunne and Brown, 1980; Highsmith, 1982; Fong and Lirman, 1995). Post-storm survival of unattached fragments generated by hurricanes has been reported as both size-dependent (Highsmith *et al.*, 1980) and size-independent (Lirman and Fong, 1997; Lirman, 2000). Broken fragments of *A. palmata* have been translocated and reattached in several restoration projects (Bruckner and Bruckner, 2001; Garrison and Ward, 2008; Williams and Miller, 2010; Forrester *et al.*, 2011), and larger fragments are reported to survive better than smaller ones (Bruckner and Bruckner, 2001, Garrison and Ward, 2008).

METHODS

Study sites and transplanting methods

This study used 832 storm-generated *A. palmata* fragments that were transplanted to a restoration

site (0.4–1.6 m deep) on the leeward southern side of Guana Island, British Virgin Islands (18°29'N, 64°35'W). The fragments were collected from three source populations, all within 4 km of the restoration site. Conditions at the study sites are described in detail elsewhere (Forrester *et al.*, 2012). Groups of fragments were transplanted in July–August each year from 2005–2011 (Table 1).

Each year, student and volunteer divers searched for fragments at the source sites, and brought all fragments they found to the surface, except fragments that showed visual symptoms of disease (Williams *et al.*, 2006). Because all undiseased fragments encountered were collected, the distribution of fragment sizes should be representative of that present in the area. Once on the boat, fragments were submerged in bins of fresh seawater. Corals were then taken to the restoration site and transplanted within 2.5 h of collection (for further details see Forrester *et al.*, 2011).

At the restoration site, divers attached fragments to the reef using one of three methods: (1) nylon cable ties; (2) marine epoxy; or (3) hydrostatic cement. In a previous analysis, these three attachment methods were shown to have no effect on the growth and survival of the transplanted fragments (Forrester *et al.*, 2011). An independent study of *A. palmata* also found no differences between cable tied and epoxied fragments (Williams and Miller, 2010). To facilitate recognition and analysis of fragments over time, the location of fragments was mapped and a numbered tag was attached to the reef adjacent to each fragment.

Monitoring fragments

Fragments were monitored at intervals after transplanting. Each cohort (fragments transplanted

Table 1. Initial sizes (surface area of live tissue in cm²) of coral fragments, grouped by year transplanted. Displayed are number of corals transplanted (*n*), mean ($\pm$ SE), minimum and maximum sizes

Year transplanted	<i>n</i>	Mean	($\pm$ SE)	Min.	Max.
2005	35	198	(37)	7	961
2006	19	78	(17)	2	300
2007	88	117	(16)	8	950
2008	254	106	(17)	15	823
2009	44	87	(22)	2	561
2010	135	79	(10)	4	1004
2011	257	113	(10)	12	1015

in the same year) was monitored 3 and 12 months after transplanting, and then annually thereafter. The size of each coral fragment was estimated within a few days of transplanting and during all subsequent surveys. Colony size was measured as the surface area of live tissue (SAL) and estimated using two methods. For small colonies with simple shapes, the surface area was traced from digital photographs of each surface using image analysis software (ImageJ) (Bythell *et al.*, 2001; Abramoff *et al.*, 2004). For large colonies, with more complex three-dimensional branching morphology, SAL was estimated from linear dimensions. SAL was estimated as $[(L + W + H)/3]^2$, where L, H and W are colony length, width and height respectively (Williams *et al.*, 2008).

Analysis of survival

Differences in survival among years were tested to assess whether interannual differences in conditions influenced the success of transplanting. As a simple means of testing for a difference in survival among cohorts, a χ^2 test was used to compare the observed number of fragments that survived their first year against the number expected under the null hypothesis of equal survival in all years (Sokal and Rohlf, 1995). The range of fragment sizes used was fairly consistent each year, and the mean initial size of fragments did not differ among cohorts (ANOVA $F_{6,825} = 1.72$, $P = 0.12$; Table 1), so this analysis should be unaffected by bias in fragment sizes.

Additional tests examined the effect of initial fragment size on survival and whether survival was a function of time elapsed since transplanting. Data from all cohorts were pooled for this analysis, which is justifiable because the range of fragment sizes used was fairly consistent across cohorts (discussed above). Pooling cohorts permitted analysis of a large sample of fragments and allowed tracking of early cohorts for several years. For this analysis, simple parametric survival models appropriate for censored (i.e. incomplete) data on survival times were used (Lee, 1992; Kleinbaum and Klein, 2011). Survival times are uncertain in field studies because the exact timing of death is almost never observed. Regularly monitoring the fragments yielded estimates of survival time (the time elapsed

between transplanting and death) that were either interval-censored (when a fragment died between two censuses) or right-censored (the fragment was still alive at the end of the study) (Lee, 1992).

Estimates of survival time are conservative because any coral that could not be positively identified was assumed to have died. There were no sexually produced *A. palmata* recruits at the restoration site; therefore transplants could not be confused with natural colonies. By the end of the study, however, there were 35 fragments at the site that could not be positively identified because they lacked a tag and/or were not in the location of an original fragment. Based on their locations and morphology, these fragments probably originated from *A. palmata* transplanted early in the study that were damaged by storms or boats, and then subsequently reattached to form a new colony.

To isolate the effect of time elapsed since transplanting, three models for the time dependence of survival were compared. The null exponential model assumes the instantaneous mortality rate is constant. The Weibull model allows the instantaneous mortality rate to either rise or decline monotonically as a function of time, whereas the log-logistic model allows the rate to either rise or decline over time, or to rise to a maximum then subsequently decline (i.e. a 'hump' shaped pattern). A parameter was then added to these models to test for the effect of fragment size on survival.

The fit of data to each model was assessed by inspecting plots of the quantiles of the fitted distribution versus observed cumulative deaths, Cox–Snell residual plots (estimated cumulative mortality), and probability plots (failure times versus the quantiles of the chosen distribution) (Davison and Tsai, 1992; Waller and Turnbull, 1992; Cohen, 1995). Relative model fit was also compared using Akaike's Information Criterion, corrected for small samples (AICc), using the convention that models differ in fit when their AICc values differ by more than 7 (Burnham and Anderson, 2002).

Analysis of change in colony size

To assess the effect of initial fragment size on the net change in colony size due to the combined

effects of loss (partial mortality) and gain (clonal reproduction) of polyps, the change in SAL was modelled as a function of initial SAL using linear regression. Two analyses were performed in which the response variables were growth over 12 months and growth over 24 months respectively, because large sample sizes were available for these time intervals ($n = 653$ and 238 respectively).

RESULTS

Analysis of survival

There were significant differences in first-year survival among cohorts ($\chi^2 = 41.9$, $df = 1$, $P < 0.0001$). The percentage of transplants surviving the first year was lower for the 2007 and 2010 cohorts than for corals transplanted in other years (Table 2).

When cohorts were pooled, both survival models allowing the instantaneous mortality rate to vary as a function of time since transplanting (log-logistic and Weibull) fit better than the simpler model that assumes a constant mortality rate (exponential) (Table 3). Of the two models allowing the mortality rate to vary over time, AICc values indicate that the log-logistic model fits better than Weibull (Table 3). This ranking of relative model fit based on AICc values was also consistent with inspection of diagnostic plots for the three models (plots not shown).

Although the log-logistic allows for complex patterns of survival over time, the mortality functions for both the Weibull and log-logistic models indicated a simple decelerating mortality rate over time. Under both models, the mortality rate was high in the first 12 months after transplanting and then relatively constant thereafter

(plot for best-fitting log-logistic function shown in Figure 1). For comparison with the data on percentage survival among cohorts, the instantaneous mortality rates were converted to annual percentage survival. This conversion requires the simplifying assumption that instantaneous mortality is constant over time (and high) for year 1, and then constant (but lower) thereafter (Figure 1). Under this simplifying assumption, roughly 56% of fragments are expected to remain alive a year after transplanting but annual survival increases to 68% for all subsequent years.

Acropora palmata fragments found at the source sites varied widely in initial size (2–1016 cm²), but most fragments were small (mean = 108 cm²) and 50% of the fragments were between 26 and 136 cm² (Figure 2). For all three survival models evaluated, adding a term for the effect of initial fragment size improved model fit relative to the comparable model lacking this covariate (Table 3). The parameter for fragment size was always positive and significantly different from zero (Table 3), indicating that all models predict increased survival of larger fragments. Adding a term for fragment size to the survival models did not alter the relative fit of the three models, and both parameters of the Weibull and log-logistic models were still statistically significant (Table 3). This indicates separate effects both of time since transplanting and fragment size on survival, i.e. poor initial survival was not simply a by-product of that fact that fragments tended to be smaller when transplanted than afterwards.

To illustrate the effect of fragment size on survival, the best-fitting model (log-logistic) was used to predict survival over time for three representative sizes of coral: 10 cm², 100 cm², and 1000 cm² (sizes close to the smallest, average and largest fragments found at the study site) (Figure 3). This plot shows a modest increase in expected survival as coral size increases from 10–100 cm², but a dramatic improvement for 1000 cm² corals. For example, 10% of the smallest corals are expected to remain alive after 7 years. For average sized corals, this figure increases to 19%, but 67% of the largest corals remain alive after 7 years.

Table 2. Initial survival of corals transplanted in different years

Year transplanted	Percentage alive after 12 months
2005	85
2006	58
2007	21*
2008	50
2009	68
2010	30*
2011	70

*Indicates years when major storms affected the site.

Table 3. Estimates of model fit (AICc) and parameter estimates ($\pm$ 95% CI) for survival models fitted to the data. Three survival models were compared: exponential, Weibull, and log-logistic. Models were first fitted without (a) and then with (b) a term for initial coral size as a covariate. The meaning of the parameters $B1$ and $B2$ is model-specific, as follows: for exponential $B1$ = scale; for Weibull $B1$ = shape and $B2$ = scale; for log-logistic $B1$ = scale and $B2$ = location

(a) Models with no covariates				
Survival model	AICc	$B1 \pm (95\% \text{ CI})$	$B2 \pm (95\% \text{ CI})$	
Exponential	1848	$19.61 \pm (17.93\text{--}21.41)$		
Weibull	1784	$0.74 \pm (0.70\text{--}0.77)$	$21.0 \pm (18.9\text{--}23.6)$	
Log-logistic	1764	$1.01 \pm (0.95\text{--}1.08)$	$2.52 \pm (2.36\text{--}2.58)$	
(b) Models with term for initial coral size				
Survival model	AICc	$B1 \pm (95\% \text{ CI})$	$B2 \pm (95\% \text{ CI})$	Initial size $\pm (95\% \text{ CI})$
Exponential	1792	$15.51 \pm (12.77\text{--}18.39)$		$0.002 \pm (0.001\text{--}0.0035)$
Weibull	1740	$1.35 \pm (1.28\text{--}1.39)$	$2.70 \pm (2.65\text{--}2.77)$	$0.003 \pm (0.002\text{--}0.004)$
Log-logistic	1718	$0.99 \pm (0.98\text{--}1.02)$	$2.10 \pm (2.04\text{--}2.17)$	$0.003 \pm (0.002\text{--}0.004)$

Effects of initial colony size on colony growth

Considering only surviving colonies, the average fragment shrank slightly in size in the first 3 months after being transplanted (Figure 4). Subsequently, surviving fragments tended to progressively increase in size, albeit with substantial variation around the trend (Figure 4).

Linear regressions for survivors only, showed that the relationship between initial colony size (SAL in cm^2) and change in colony size was statistically significant for change over 12 months ($r^2 = 0.06$, $F_{1,385} = 25.6$, $P < 0.0002$), but not for change over 24 months ($r^2 = 0.02$, $F_{1,143} = 2.79$, $P = 0.196$). Although the relationship was statistically significant after 12 months, initial colony size (SAL in cm^2) explained only a trivial percentage of variation in colony growth in the first 12 and 24 months after transplanting (6% and 2% respectively; Figure 5).

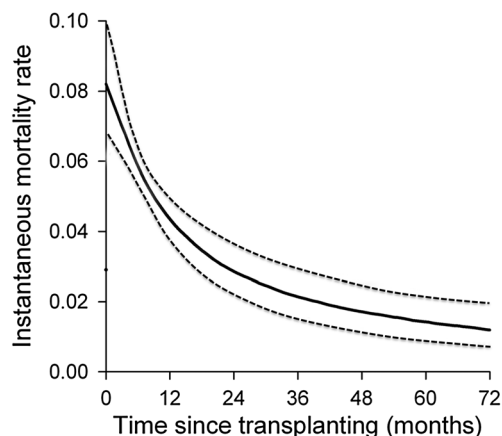


Figure 1. Instantaneous mortality rate after transplanting for the best fitting log-logistic survival model. Dotted lines are 95% confidence intervals.

DISCUSSION

Effects on survival of time elapsed since transplanting and year transplanted

For any fragment loose on the reef, survival depends on remaining at a favourable location long enough for the colony to reattach itself to the reef (Tunncliffe, 1981; Fong and Lirman, 1995; Smith and Hughes, 1999). For this reason, fragments reattached manually for restoration have an initial advantage over loose fragments (Lindahl, 2003; Forrester *et al.*, 2011). For example, at the sites from which *A. palmata* was collected, only 2% (2 out of 102) of fragments left loose as controls remained alive after 12 months, so their survival was far worse than transplanted fragments (Forrester, unpublished data).

Survival in the first 12 months after transplanting varied considerably among the 7 years when corals

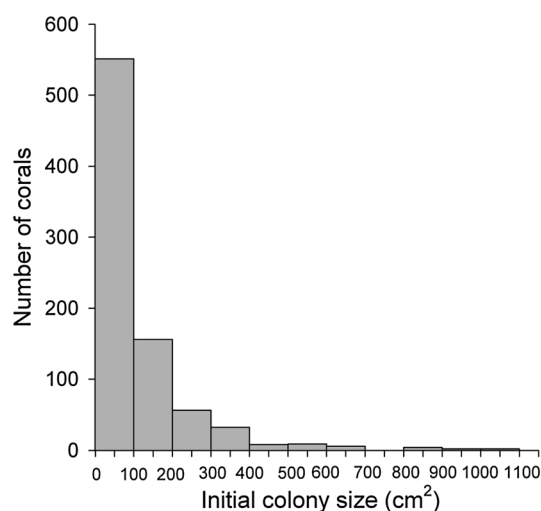


Figure 2. Size distribution of coral fragments encountered at source sites and used for transplanting.

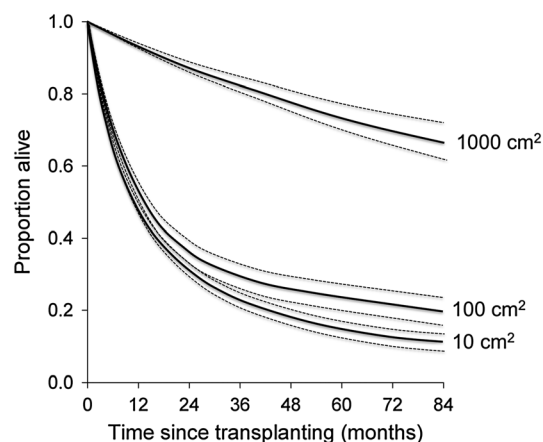


Figure 3. Survival as a function of time since transplanting for three representative sizes of transplanted corals. Sizes were selected to represent corals close to the smallest, average, and largest coral fragments found naturally at the site: 10, 100, and 1000 cm² respectively. Solid lines are best-fit estimates from the log-logistic survival function and dotted lines are 95% confidence intervals.

were transplanted (2005–2011). These interannual differences are at least partly due to the effects of storms, whose ability to inflict severe episodic mortality on *A. palmata* is well documented (Highsmith *et al.*, 1980; Woodley *et al.*, 1981; Rogers *et al.*, 1982). In the 2 years when survival to 12 months was lowest (2007 and 2010), storms caused visible damage to the benthos and reef structure at the site (a winter storm on 19–20 March 2008, and hurricane Earl on 30 August 2010, respectively). Interestingly, survival was highest in 2005 even though a warming event caused mass bleaching throughout the Caribbean in autumn of that year. In addition, pooling data across cohorts showed that survival was consistently lower in the 12 months after transplanting than thereafter. Low initial survival may be attributed partly to stress induced by the transplanting process (Yap and Gomez, 1985; Plucer-Rosario and Randall, 1987; Yap *et al.*, 1992; Forrester *et al.*, 2011). Transplanting stress is further indicated by the fact that surviving fragments tended to decrease in size in the first 3 months after transplanting, but progressively increased in size thereafter. Another probable cause of low initial survival is failure of the ties/epoxy/cement used to reattach the fragment (Borneman and Lowrie, 2001; Bruckner and Bruckner, 2001). Deaths from attachment failure

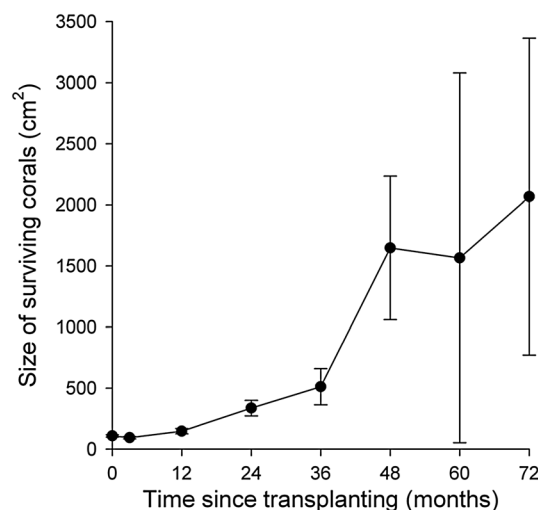


Figure 4. Mean colony size (mean SAL in cm² ± 95% CI) of corals as a function of time since transplanting. Only corals that remained alive at each time were plotted.

should decline rapidly with elapsed time as fragments grow and form their own connection with the substratum (Guest *et al.*, 2011) and it was noted during monitoring visits that, after 12 months, the tissues of virtually all surviving *A. palmata* fragments had grown to contact the reef.

Effects of initial colony size on growth and survival

Fragment survival improved with increasing colony size with the very largest fragments surviving far better than small and average-sized fragments. These results are consistent with the analysis of the survival of *A. palmata* fragments reattached to the reef with wire after a ship grounding in Puerto Rico (Bruckner and Bruckner, 2001). Survivors after 2 years (mean length = 70 cm) were significantly larger than fatalities (mean length = 52 cm), indicating better survival of larger fragments. These fragments generated by ship grounding (15–304 cm long) were generally larger than the (presumably) storm-generated fragments transplanted in the study (where colony size was expressed as surface area but, for comparison, fragments ranged from 4–105 cm long). Okubo and colleagues also reported size-dependent survival of two Pacific species of *Acropora*. They pruned small fragments from natural colonies of *A. formosa* (5, 10 and 20 cm in length) and *A. nasuta*

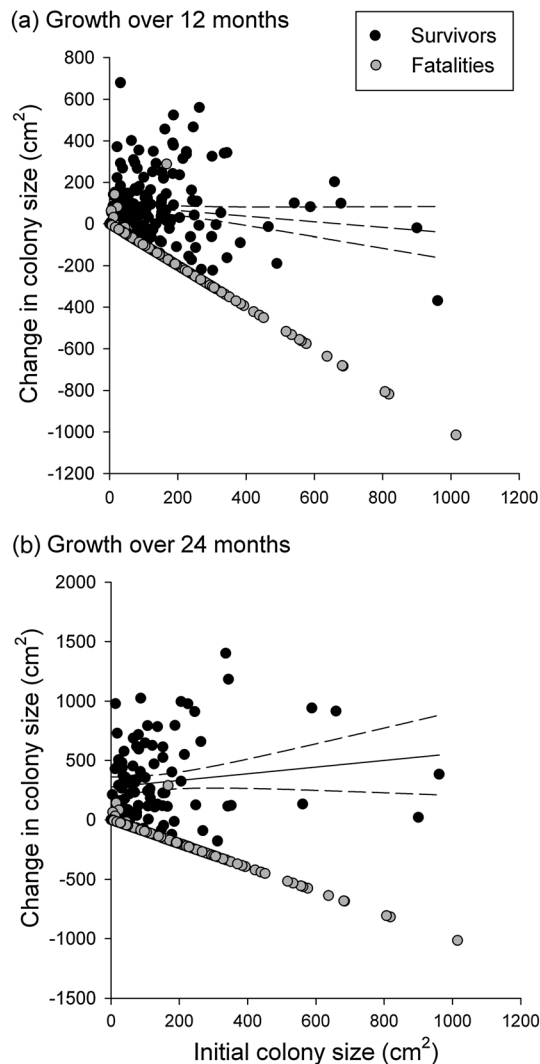


Figure 5. Change in colony size (SAL in cm^2) as a function of initial size. Data are plotted for growth over the first 12 months (a) and 24 months (b) since transplanting. Corals that survived the interval and those that did not are plotted separately, and a linear regression (solid line) with 95%CI (dotted lines) is fitted through data for survivors only.

(5 and 10 cm in length) and cable-tied them to nails that had been hammered part-way into the reef. Larger fragments of both species survived better than smaller ones after 1 year and 3 years respectively (Okubo *et al.*, 2007, 2009). *Acropora* fragments affixed to the substratum for restoration thus appear to consistently display size-dependent survival.

Survival is also commonly a positive function of size for natural coral colonies (Connell, 1973; Hughes and Jackson, 1985; Vermeij and Sandin, 2008). The survival of transplanted fragments is likely to differ from that of natural colonies for at

least three reasons: (1) transplanting stress; (2) failure of the manual reattachment process; and (3) microhabitat differences - the specific locations chosen by restoration practitioners to attach fragments are likely to be different from locations selected by sexual coral recruits (Hughes, 1984). Nonetheless, two mechanisms for increased survival with larger size should apply equally to natural colonies and transplants: (1) a reduced probability that predators, disease or abrasion can kill all polyps in a colony; and (2) an improved ability to mobilize resources from unaffected polyps (Pearse and Muscatine, 1971; Hughes, 1984).

To the best of our knowledge, the only direct comparison between natural colonies and manually reattached fragments was made by Garrison and Ward (2008). They found no difference in survival of natural colonies and manually reattached fragments of *Acropora cervicornis* and *Porites porites*, though their samples were small enough that subtle differences in survival might not be detectable ($n=15$ transplants and $n=15$ natural colonies per species). They were able to compare a larger sample of *A. palmata* ($n=30$ transplants and $n=45$ natural colonies) and found that natural colonies survived better than transplanted ones. Their natural colonies were, however, also larger on average than transplanted colonies so better survival may be a function of larger size. Further direct comparisons between natural colonies and manually reattached fragments would thus be valuable to test whether size-based demographic models developed for natural colonies (Hughes, 1984; Hughes and Jackson, 1985) might be modified to predict the fate of transplanted fragments.

Although survival was size-dependent, the change in size of surviving *A. palmata* colonies was independent of their initial size. In contrast, experimentally created fragments of *Acropora nasuta*, *Porites rus* and *Porites cylindrica*, all displayed a positive relationship between size and colony growth after attachment to the reef (Yap *et al.*, 1998; Okubo *et al.*, 2009). For *A. nasuta*, this represented a trade-off between growth and reproduction, smaller fragments grew faster than larger ones at the expense of oocyte production (Okubo *et al.*, 2009). *Acropora palmata* reproduces

sexually once per year, but it was not possible to monitor spawning during this study. Nonetheless, size-independence of colony growth in *A. palmata* presumably reflects the fact that, like all modular colonial organisms, coral colonies can shrink as well as grow over time and at the study sites, surviving colonies were just as likely to lose as they were to gain polyps in any given time interval (Garrison and Ward, 2008). This matches the pattern of growth reported for natural colonies *Agaricia agaricites*, whose colonies frequently decreased as well as increased in size between monitoring events (Hughes, 1984).

Implications for restoration

For coral fragments transplanted and reattached to restore reefs, one useful measure of their success is whether they grow and survive as well as natural colonies. Although there were obvious impacts of the transplanting process, in terms of reduced colony growth in the first 3 months and reduced survival in the first 12 months after transplanting, it is encouraging that transplanting effects were of no greater magnitude than year-to-year differences in survival from natural causes (e.g. storms). Beyond the first year, fragment survival was reasonably high and, although differences in methods preclude an exact comparison, long-term survival of the larger transplanted fragments appears roughly equivalent to the survival of large natural *A. palmata* colonies in Florida (Williams and Miller, 2012). For practitioners, these results suggest that demographic monitoring of restoration projects should focus on the first year after transplanting.

Students and volunteers were used to transplant most fragments, rather than professional scientists, in order to simulate a level of care and expertise that might be expected in volunteer-based restoration projects. This restoration method depends on locating a donor population of *A. palmata* with fragments suitable for transplanting. Every year in the study area, it was possible to find enough fragments in a few days of searching. Before starting the transplanting process, students and volunteers received training for 1 or 2 days on how to attach fragments to the reef, and on

selecting suitable locations for attachment. The overall process of finding and transplanting fragments thus took less than a week, and so is feasible for projects using volunteers that have limited time to invest in conservation work. Simple, direct transplanting of *A. palmata* thus shows promise as a method usable by practitioners to restore local populations of endangered *A. palmata* on Caribbean reefs.

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